Power, phasors and complex power

Answers and worked solutions · Day 01 · Sat 24 Oct · pack M1

5. Practice

From your course files

Power Circuit Tool WS1 Q1 (workshop).

Hint: $P = 100$ W, $Q = 100$ VAr, $V = 240$ V.

a. The tool shows $v(t) = \sqrt{2} \times 240 \sin(2\pi 50 t)$, so $V_{rms} =$ **240 V** and $V_{pk} = \sqrt{2} \times 240 =$ **339.4 V**. $S = \sqrt{100^2 + 100^2} = 141.4$ VA, so $I_{rms} = S/V = 141.4/240 =$ **0.589 A** and $I_{pk} = \sqrt{2} \times 0.589 =$ **0.833 A**.

b. $Q = +100$ VAr: the current **lags** and the pf is **lagging** (inductive load absorbs Q). $Q = 0$: the current is **in phase** and the pf is **unity**. $Q = -100$ VAr: the current **leads** and the pf is **leading** (capacitive load generates Q).

These match the official answers.

Power Circuit Tool WS1 Q2 (workshop).

a. **Anticlockwise** (the standard direction for phasors, at $\omega = 2\pi 50$ rad/s).

b. The **y-axis** component. The tool's waveforms are written with $\sin$, so the vertical projection of each rotating phasor gives the instantaneous value.

c. With V as the reference at $0°$, look at the angle of the current phasor. Below the voltage phasor (clockwise from it, e.g. $-45°$ for $Q = +100$ VAr) means **lagging**. On top of it means **in phase**. Above it (anticlockwise, $+45°$ for $Q = -100$ VAr) means **leading**.

Power Circuit Tool WS1 Q3 (workshop).

a. $p(t)$ repeats every $T = 10$ ms, so $f = 1/T =$ **100 Hz**, twice the 50 Hz supply frequency.

b. Whenever $v(t) = 0$ or $i(t) = 0$.

(c)(i) **No change.** The average stays at $P = 100$ W for all three values of Q.

ii. $p(t) = P + S\cos(2\omega t + \phi)$, so its minimum is $P - S$. For $Q = 0$: $S = 100$ VA and the minimum is **0 W**. For $Q = \pm 100$ VAr: $S = 141.4$ VA and the minimum is

$100 - 141.4 = -41.4$ **W**. The official sheet reads about −41.2 W off the graph, so the difference is only graph-reading accuracy.

PQ 1.14 (practice questions).

Hint: $\omega = 2\pi n/60$ every time, then $P = \tau\omega$.

a. $\omega = 2\pi(2910)/60 = 304.7$ rad/s, so $\tau = 7500/304.7 =$ **24.61 Nm**.

b. $\omega = 2\pi(50\,000)/60 = 5236$ rad/s, so $P = 4.25 \times 5236 =$ **22.25 kW**.

c. $\omega = P/\tau = 30 \times 10^6/191 \times 10^3 = 157.1$ rad/s, so $n = 60\omega/2\pi =$ **1500 rpm**.

These match the official answers.

PQ 1.21 (practice questions).

Hint: $P_{OUT} = \tau\omega$.

a. $P_{IN} = V_1 I_1 \cos\phi = 110 \times 15 \times 0.8 =$ **1.320 kW**.

b. $\omega = 2\pi(1750)/60 = 183.3$ rad/s, so $P_{OUT} = 6 \times 183.3 =$ **1.100 kW**.

c. $\eta = 1100/1320 =$ **83.30 %**.

d. $P_{LOSS} = 1320 - 1100 =$ **220 W**.

These match the official answers.

PQ 1.22 (practice questions).

Hint: the 240 V is the **phase** voltage, so $I = P_{IN}/(3V_1\cos\phi)$.

a. $\omega = 2\pi(650)/60 = 68.07$ rad/s, so $P_{OUT} = 63 \times 68.07 =$ **4.288 kW**.

b. $P_{IN} = 4288/0.88 =$ **4.873 kW**.

c. $P_{LOSS} = 4873 - 4288 =$ **585 W**.

d. $I_1 = \dfrac{4873}{3 \times 240 \times 0.85} =$ **7.962 A**.

These match the official answers.

PQ 2.6 (practice questions).

Hint: maximum power is at $R_L = R_S$.

a. No current, so no drop: $V_{oc} = E =$ **12.5 V**.

b. $I_{sc} = 12.5/0.02 =$ **625 A**, and $P = I^2 R_S = 625^2 \times 0.02 =$ **7.813 kW**. (Don't short-circuit batteries: this heats them very quickly.)

c. $I = 12.5/(0.02 + 1) = 12.25$ A and $V = IR_L = 12.25$ V. The official sheet gives **12.26 V, 12.26 A**; the exact value is 12.2549, so the difference is rounding.

d. $I = (E - V)/R_S = (12.5 - 8)/0.02 =$ **225 A**, and $P = VI = 8 \times 225 =$ **1.800 kW**.

e. At $R_L = R_S = 0.02\ \Omega$: $V = 6.25$ V, $I = 12.5/0.04 = 312.5$ A, so $P_{max} = E^2/(4R_S) = 12.5^2/0.08 =$ **1.953 kW**.

PQ 2.15 (practice questions, do (f)–(h)).

Hint: in (h), add the currents.

Quick answers for (a)–(e): $24\angle 0°$, $10\angle 0°$, $7.059\angle 0°$, $34\angle 0°$ and $10\angle -90°$ A.

f. $Z = 24 + j24 = 33.94\angle 45°\ \Omega$, so $I = 240/33.94\angle 45° =$ **7.071∠−45° A**.

g. Add the branch currents: $240/24 + 240/(j24) = 10 - j10 =$ **14.14∠−45° A**.

h. $I = 24\angle 0° + 7.071\angle -45° = 24 + 5 - j5 = 29 - j5 =$ **29.43∠−9.782° A**.

These match the official answers.

PQ 2.17 (practice questions).

Hint: in (f)(iii), use V^2/R.

a. $Z = \dfrac{25 \times j125}{25 + j125} =$ **24.51∠11.31° Ω**.

b. $V_1 =$ **240∠0° V**.

c. $I_1 = V_1/Z =$ **9.792∠−11.31° A**.

d. pf $= \cos(11.31°) =$ **0.9806 lag**.

e. $S_1 = V_1 I_1^* = 240 \times 9.792\angle +11.31° =$ **2350∠11.31° VA = 2304 + j460.9 VA**.

f. i. $\mathrm{Re}(S_1) = 2304$ W. (ii) $V_1 I_1 \cos\phi = 240 \times 9.792 \times 0.9806 = 2304$ W. (iii) $V^2/R = 240^2/25 =$ **2.304 kW**. You can't use $I_1^2 R$, because I_1 is the line current, and only part of it flows through the resistor.

g. $Q = \mathrm{Im}(S_1) =$ **+460.9 VAr** (check: $V^2/X = 240^2/125 = 460.8$ VAr).

h. $S = $ **2350 VA**.

i. $2304^2 + 460.9^2 = 2350^2$. The triangle has P = 2304 W along the horizontal, Q = +460.9 VAr upwards, and the angle between S and P is 11.31°.

j. V_1 is along 0°, and I_1 (9.792 A) lags it by 11.31°.

These match the official answers.

PQ 2.18 (practice questions).

Hint: the voltages add at 90°.

a. V_R and V_L are perpendicular, so $V_L = \sqrt{200^2 - 150^2} =$ **132.3 V**.

b. $\phi = -\cos^{-1}(V_R/V_1) = -\cos^{-1}(0.75) =$ **−41.41°** (negative because the current lags). pf = **0.75 lag**.

c. $S = V_1 I_1 = 200 \times 10 =$ **2000 VA**, $P = S\cos\phi =$ **1500 W**, $Q = S\sin(-\phi) =$ **+1323 VAr**.

d. Resistor: **1500 W, 0 VAr**.

e. Inductor: **0 W, +1323 VAr**.

f. $R = P/I^2 = 1500/100 =$ **15 Ω** and $X = Q/I^2 = 1323/100 =$ **13.23 Ω** (reactance is a scalar number).

These match the official answers.

PQ 2.19 (practice questions).

Hint: Q is negative (capacitor).

a. All the real power is in R, which has the full V_1 across it: $R = V^2/P = 140^2/500 =$ **39.2 Ω**.

b. $S = V_1 I_1 = 140 \times 15 =$ **2100 VA**. $Q = -\sqrt{S^2 - P^2} = -\sqrt{2100^2 - 500^2} =$ **−2040 VAr** (negative because of the capacitor).

c. $\phi = +\cos^{-1}(500/2100) =$ **+76.23°**, so pf = **0.2381 leading**.

d. $P/S = 500/2100 = 0.2381$ leading, the same.

e. $X = V^2/Q = 140^2/(-2040) =$ **−9.608 Ω**. (Using the unrounded Q gives −9.610 Ω.)

f. V_1 at 0°. I_R is in phase with V_1. I_C points straight up ($+90^\circ$). $I_1 = I_R + I_C$ is the diagonal, $\phi = 76.23^\circ$ above V_1.

g. P = 500 W horizontal, Q = −2040 VAr downwards, S = 2100 VA as the hypotenuse, with ϕ between S and P.

These match the official answers.

Jun23 Q1a (past exam, 5 marks).

Hint: Playbook A.

(i) On open circuit $I = 0$, so there is no drop across R_S: $E = V_{oc} = \mathbf{13.8\ V}$.

(ii) With the starter motor, $I = 300$ A and $V = 9.5$ V:

$$R_S = \frac{E - V}{I} = \frac{13.8 - 9.5}{300} = \mathbf{0.01433\ \Omega}$$

$$P_{R_S} = I^2 R_S = 300^2 \times 0.01433 = \mathbf{1290\ W}, \qquad P_{motor} = VI = 9.5 \times 300 = \mathbf{2850\ W}$$

Check: $EI = 13.8 \times 300 = 4140$ W, which equals 1290 + 2850.

(iii) $V = E - IR_S = 13.8 - 20 \times 0.01433 = \mathbf{13.51\ V}$.

All of these match the official answers.

Jun23 Q1c (past exam, 9 marks).

Hint: Playbook C. The element values are in the course figure above: $R = 10\ \Omega$, $X_L = 30\ \Omega$ and $X_C = 50\ \Omega$. The given current is $I_1 = 13\angle{+20^\circ}$ A.

Circuit for Jun23 Q1c: series R = 10 Ω, jX_L = j30 Ω, −jX_C = −j50 Ω carrying I_1 = 13∠20° A (values from the course figure)

(i) Ohm's law for each element, using the same current through each:

$$V_R = I_1 R = 13\angle{20^\circ} \times 10 = \mathbf{130\angle{20^\circ}\ V}$$

$$V_L = I_1(jX_L) = 13\angle{20^\circ} \times 30\angle{90^\circ} = \mathbf{390\angle{110^\circ}\ V}$$

$$V_C = I_1(-jX_C) = 13\angle{20^\circ} \times 50\angle{-90^\circ} = \mathbf{650\angle{-70^\circ}\ V}$$

The official sheet's "130 120 V" is a text-extraction artefact of $130\angle{20^\circ}$. V_C (650 V) is larger than the supply voltage. This is normal in a series RLC, because V_L and V_C partly cancel.

(ii) $Z = 10 + j30 - j50 = 10 - j20 = \mathbf{22.36\angle{-63.43^\circ}\ \Omega}$. This is capacitive overall.

$$V_1 = I_1 Z = 13\angle{20^\circ} \times 22.36\angle{-63.43^\circ} = \mathbf{290.7\angle{-43.43^\circ}\ V}$$

(iii) $I_1^* = \mathbf{13\angle{-20^\circ}\ A}$.

$$S = V_1 I_1^* = 290.7\angle -43.43^\circ \times 13\angle -20^\circ = 3779\angle -63.43^\circ = \mathbf{1690 - j3380\ VA}$$

So $P =$ **1690 W** and $Q =$ **−3380 VAr**. The source receives reactive power, because the load is capacitive and the current leads.

Check: $I^2R = 169 \times 10 = 1690$ W and $I^2(X_L - X_C) = 169 \times (-20) = -3380$ VAr. The official answer of −3379 VAr is the same value with rounded intermediate steps.

Jun24 Q1a(ii) (past exam, 2 marks).

$P_{OUT} = 1000 - 300 =$ **700 W** and $\eta = 700/1000 =$ **70 %**. More loss for the same input means η **decreases**. More input for the same loss means η **increases**, because the loss becomes a smaller fraction of the input.

These match the official answers.

Jun24 Q1b (past exam, 3 marks).

Hint: Playbook B. A 3ph nameplate gives line values.

i. $S = \sqrt{3} V_L I_L = \sqrt{3} \times 415 \times 6.8 = 4888$ VA, so pf $= P/S = 4000/4888 =$ **0.8184 lag**.

ii. $\omega = 2\pi(1420)/60 = 148.7$ rad/s, so $P_{OUT} = \tau\omega = 23 \times 148.7 =$ **3.420 kW** and $\eta = 3420/4000 =$ **85.50 %**.

The “4 pole, 50 Hz” information is not needed here. It gives $n_s = 1500$ rpm, which you will use in Ch7.

These match the official answers.

Jun24 Q1c (past exam, 5 marks).

Hint: Playbook D. From the course figure, $R = 100\ \Omega$ and $X_L = 40\ \Omega$.

Circuit for Jun24 Q1c: series R = 100 Ω and jX_L = j40 Ω carrying 2 A (values from the course figure)

i. The same current $I_1 = 2$ A flows through both elements. Resistor: $P = I^2R =$ **400 W**, $Q =$ **0 VAr**. Inductor: $P =$ **0 W**, $Q = I^2X_L =$ **160 VAr**.

ii. V_R is in phase with I_1, and V_L leads it by 90°, so they are perpendicular:

$$V_1 = \sqrt{V_R^2 + V_L^2} = \sqrt{50^2 + 20^2} = \mathbf{53.85\ V}$$

Adding the magnitudes (70 V) is the classic wrong answer.

These match the official answers.

New questions

N1. Single-phase pump motor [5 marks].

a. $P_{IN} = V_1 I_1 \cos\phi = 230 \times 6.2 \times 0.78 =$ **1112 W**. $\eta = 950/1112 =$ **85.41 %** (the loss is 162.3 W).

b. $\omega = 2\pi(2860)/60 = 299.5$ rad/s, so $\tau = 950/299.5 =$ **3.172 Nm**.

c. $S = V_1 I_1 = 230 \times 6.2 =$ **1426 VA**. $Q = S\sin(-\phi)$ with $\phi = -\arccos 0.78 = -38.74°$, so $Q = 1426 \times \sin(38.74°) =$ **+892.4 VAr**. It is positive because the motor is inductive.

N2. Series RC [6 marks].

a. $X_C = 1/(2\pi \cdot 50 \cdot 100 \times 10^{-6}) = 31.83\ \Omega$. $Z = 30 - j31.83 =$ **43.74∠−46.70° Ω**.

b. $I_1 = 240\angle 0°/43.74\angle -46.70° =$ **5.487∠+46.70° A**. This leads, as a capacitive circuit should.

$V_R = I_1 R =$ **164.6∠46.70° V**. $V_C = I_1(-jX_C) = 5.487\angle 46.70° \times 31.83\angle -90° =$ **174.6∠−43.30° V**.

$\sqrt{164.6^2 + 174.6^2} = 240.0$ V, which equals V_1.

c. $S = V_1 I_1^* = 240 \times 5.487\angle -46.70° = 1317\angle -46.70° =$ **903.2 − j958.3 VA**. So $P = 903.2$ W, $Q = -958.3$ VAr and pf $= \cos(46.70°) =$ **0.6859 lead**.

Check: $I^2 R = 5.487^2 \times 30 = 903.2$ W.

N3. One measurement, two models [6 marks].

a. $S = 230 \times 5 =$ **1150 VA**. $Q = +\sqrt{1150^2 - 900^2} =$ **+715.9 VAr** (positive because the load is inductive). pf $= 900/1150 =$ **0.7826 lag**.

b. $R_S = P/I^2 = 900/25 =$ **36.0 Ω**, $X_S = Q/I^2 = 715.9/25 =$ **28.64 Ω**, and $L_S = 28.64/(2\pi \cdot 50) =$ **91.15 mH**.

c. $R_P = V^2/P = 230^2/900 =$ **58.78 Ω**, $X_P = V^2/Q = 230^2/715.9 =$ **73.89 Ω**, and $L_P =$ **235.2 mH**.

Check: converting $36 + j28.64$ to its parallel equivalent through $1/Z$ gives the same 58.78 Ω and 73.89 Ω.

N4. Two loads on one supply (chained) [8 marks].

a. $P_A = 1500/0.80 =$ **1875 W**. $S_A = P_A/\text{pf} = 1875/0.75 =$ **2500 VA**.
$Q_A = S_A \sin(\arccos 0.75) =$ **+1654 VAr**. $I_A = 2500/240 =$ **10.42 A**, with angle $-41.41°$.

b. $Z_B = 40 - j30 = 50\angle -36.87°\ \Omega$, so $I_B = 240/50\angle -36.87° =$ **4.8∠+36.87° A**.

$S_B = VI_B^* = 240 \times 4.8\angle -36.87° =$ **921.6 − j691.2 VA**. Check:
$I^2R = 23.04 \times 40 = 921.6$ W.

c. $P = 1875 + 921.6 =$ **2797 W**. $Q = 1654 - 691.2 =$ **+962.4 VAr**.
$S = \sqrt{2797^2 + 962.4^2} =$ **2958 VA**. pf $= 2797/2958 =$ **0.9456 lag**.
$I = 2958/240 =$ **12.32 A**.

Check by adding the phasors: $10.42\angle -41.41° + 4.8\angle 36.87° = 12.32\angle -18.99°$ A. Note that $|I_A| + |I_B| = 15.2$ A is not the supply current. The capacitive load B partly cancels A's reactive power, which is power-factor correction in disguise (pack M2).

N5. Series RLC with a surprise [7 marks].

a. $X_L = 2\pi(50)(0.04) =$ **12.57 Ω** and $X_C = 1/(2\pi \cdot 50 \cdot 200 \times 10^{-6}) =$ **15.92 Ω**.
$Z = 8 + j(12.57 - 15.92) = 8 - j3.349 =$ **8.673∠−22.72° Ω**. It is **capacitive** overall, because $X_C > X_L$.

b. $I_1 = 230/8.673\angle -22.72° =$ **26.52∠+22.72° A**. $V_R = 212.2\angle 22.72°$ V, $V_L = 333.3\angle 112.72°$ V and $V_C = 422.1\angle -67.28°$ V. $|V_C| = 422$ V is much larger than the 230 V supply. V_L and V_C are $180°$ apart and mostly cancel, so the capacitor must be rated for the larger voltage.

c. $S = 230 \times 26.52\angle -22.72° = 6100\angle -22.72° =$ **5626 − j2355 VA**. So $P = 5626$ W, $Q = -2355$ VAr and pf $=$ **0.9224 lead**.

Check: $I^2R = 26.52^2 \times 8 = 5626$ W. The inductor absorbs $I^2X_L = +8838$ VAr and the capacitor generates $-I^2X_C = -11193$ VAr, which sum to -2355 VAr.

d. Unity pf needs $X_L = X_C$, i.e. $f_0 = \dfrac{1}{2\pi\sqrt{LC}} = \dfrac{1}{2\pi\sqrt{0.04 \times 200 \times 10^{-6}}} =$ **56.27 Hz**.

N6. Three-phase motor nameplate (brief preview of M2) [6 marks].

a. The nameplate power is the output. $\omega = 2\pi(1455)/60 = 152.4$ rad/s, so $\tau = 7500/152.4 =$ **49.22 Nm**.

b. $P_{IN} = 7500/0.88 =$ **8.523 kW**, so $P_{LOSS} =$ **1.023 kW**.

c. $I_L = P_{IN}/(\sqrt{3}V_L\,\text{pf}) = 8523/(\sqrt{3}\times 415\times 0.84) =$ **14.12 A**.
$S = 8523/0.84 = 10.15$ kVA, so $Q = S\sin(\arccos 0.84) =$ **+5.505 kVAr**.

d. Assume star, as the course does when the connection isn't stated: $V_1 = 415/\sqrt{3} =$ **239.6 V**.

6. Quick recall check

Question

1. $\sqrt{2}\times 230 = 325.3$ V.
2. 314.2 rad/s; 954.9 rpm.
3. Leading (Q < 0 means capacitive). $S = 1300$ VA, so pf $= 1200/1300 = 0.9231$ lead.
4. $\sqrt{80^2+60^2} = 100$ V, not 140 V.
5. S is a magnitude, and the two loads' complex powers point in different directions. Only P and Q are conserved, so add those and then use $S = \sqrt{P^2+Q^2}$.